Understanding Inflation in New Zealand: Trends, Dynamics, and Short-Term Forecasting

Wayne Aung

April 2026

1. Introduction

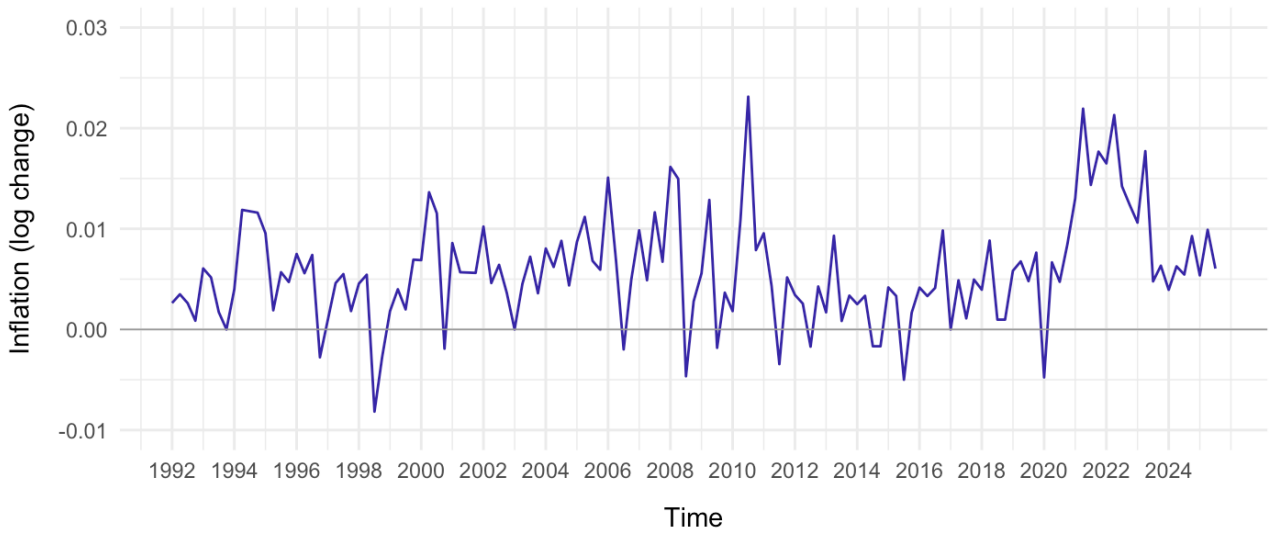
Inflation is a key macroeconomic indicator, which reflects the general price level in an economy. Historically, New Zealand experienced relatively high and persistent inflation prior to the introduction of the modern monetary policy framework, with inflation rates exceeding those of many comparable economies (McDermott & Williams, 2018). In response, the Reserve Bank of New Zealand Act 1989 introduced an inflation targeting regime in 1990. Since then, inflation has generally remained within or near the targeted band (Ball et al., 2025).

This study aims to examine the dynamics of inflation in New Zealand and to establish a baseline model for short-term forecasting. Understanding these dynamics is important for capturing temporal patterns and persistence in inflation.

Quarterly data are used in this analysis, as the official Consumer Price Index (CPI), in New Zealand is reported on a quarterly basis (Stats NZ, 2026). This makes it an appropriate frequency for analysis. The sample period is restricted to 1992–2025, to ensure a more structurally consistent monetary policy regime following the introduction of inflation targeting in the early 1990s (Buckle, 2018). Thus, the dataset used in this study includes observations up to 2025 December quarter, as it represents the most recent data available at the time of writing (April 2026).

Figure 1.

Quarterly Inflation in New Zealand, 1992–2025.



Note. Inflation is measured as the log difference of CPI.

Figure 1 illustrates quarterly inflation in New Zealand from 1992 to 2025. The series exhibits notable volatility, with several periods of prominent fluctuations. Increased volatility is observed during the late 1990s and around the 2008-2009 Global Financial Crisis, indicating the presence of macroeconomic shocks. A temporary spike is also evident around 2010-2011, followed by a period of relative stabilisation. More recently, inflation shows renewed volatility in the post-COVID period.

Despite these short-term fluctuations, the series appears to fluctuate around a relatively stable level over time. This visual pattern motivates formal testing of stationarity in the subsequent analysis.

2. ARIMA Model Specification

The Autoregressive Integrated Moving Average (ARIMA) model is a univariate time series framework commonly used in short-run forecasting. It models a variable as a function of its own past values and past error terms through autoregressive (AR) and moving average (MA) components. In its general form, the model is written as ARIMA (p, d, q), where p denotes the number of autoregressive lags, d the order of differencing required to achieve stationarity and q the number of moving average lags. ARIMA requires the series to be stationary. Non-stationary series are thus transformed through differencing (Box et al., 2016).

Given the nature of price level data, the CPI in levels exhibits a persistent upward trend and is generally non-stationary. To address this, the series is transformed using the first difference of the natural logarithm. This transformation approximates percentage changes, allowing inflation to be expressed as quarter-on-quarter growth. In addition, differencing helps remove non-stationarity associated with trends, making the series more suitable for time series modelling. The equation of the quarter-on-quarter inflation is as below:

$$\pi_t = \ln(CPI_t) - \ln(CPI_{t-1})$$

Where:

π_t = quarter-on-quarter inflation (log change) at time t

CPI_t = consumer Price Index at time t

CPI_{t-1} = consumer Price Index at time t-1

ln ( ) = natural logarithm

Accordingly, the differencing order (d=1) is implied by the log-transformation applied to the data. Hence, the general ARIMA model is specified as follows:

$$\pi_t = \alpha + \sum_{i=1}^{p} \emptyset_i \, \pi_{t-i} + \sum_{j=1}^{q} \theta_j \, \varepsilon_{t-j} + \varepsilon_t$$

Where:

α = intercept

$\emptyset_i$ = autoregressive (AR) coefficients

θ_j = moving average (MA) coefficients

ε_t = white noise error term at time t

p = number of AR lags

q = number of MA lags

3. Data and Methodology

The data used in this study were obtained from Stats NZ's *Consumer Price Index: December 2025 quarter – Index Numbers* dataset. All data preparation and analysis were conducted using R (Appendix A). The CPI series was filtered to retain the "All Groups for New Zealand" category to ensure a consistent measure of overall price movements. Missing and zero values were checked and found to be negligible for the selected sample.

A time index was constructed by converting the period variable into a date format, and the data were arranged chronologically to support the time series analysis. The dataset was then restricted to the relevant sample period (1992–2025). Subsequently, the CPI series was log-differenced to obtain quarterly inflation. As this transformation requires a lagged observation, the first available period (1992 March Quarter) is excluded, resulting in a final sample of 135 observations used for modelling.

To model the dynamics of inflation, Box–Jenkins methodology (Box et al., 2016) was followed. Stationarity of the transformed series was assessed using the Augmented Dickey-Fuller (ADF) test for the presence of a unit root (Dickey & Fuller, 1979). Autocorrelation (ACF) and partial autocorrelation (PACF) functions (Box et al., 2016) were then examined to inform model identification. Based on those patterns, several candidate ARIMA models were estimated. Model selection was based on information criteria (AIC and BIC), balancing model fit and parsimony.

The selected model was further validated through residual diagnostics, including inspection of residual autocorrelation (the Ljung-Box test). For comparison, the chosen specification was also evaluated against the model selected using the automated auto.arima() procedure. Finally, the fitted model was used to generate short-term inflation forecasts. Appendix B includes the ACF and PACF plots, along with residual diagnostic plots.

4. Results

As mentioned previously, to further test stationarity, the ADF test was conducted on the log-differenced time series which had already been transformed to reduce non-stationarity. The ADF test failed to reject the null hypothesis of a unit root at the 5% significance level ($p \approx 0.07$). This suggests that the null hypothesis of a unit root cannot be rejected at the 5% level, indicating weak evidence against non-stationarity, although rejection is achieved at the 10% level. Referring back to Figure 1, the visual inspection implied that inflation fluctuates around a relatively stable mean level over time.

Therefore, ACF and PACF plots were retrieved for inspecting AR and MA tendencies. The ACF displayed a gradual decay, while the PACF showed a prominent spike at lag 1 followed by values that are largely within the confidence bounds. This pattern is broadly consistent with a low-order autoregressive structure. However, the gradual decay in the ACF also suggests that a mixed ARMA specification may better capture the dynamics of the series.

Based on the patterns observed in the ACF and PACF plots, several candidate ARIMA models were estimated. Table 1 presents a comparison of these models using Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Lower values of these criteria indicate a better balance between model fit and parsimony (Hyndman & Athanasopoulos, 2018).

Table 1.

Comparison of candidate ARIMA models based on AIC and BIC

Model	AIC	BIC
ARIMA (1,0,0)	-1046.51	-1037.8
ARIMA (0,0,1)	-1039.61	-1030.89
ARIMA (1,0,1)	-1048.89	-1037.27
ARIMA (2,0,1)	-1047.07	-1032.54
auto.arima	-1048.89	-1037.27

Among the candidate models, ARIMA (1, 0, 1) produced the lowest AIC and BIC values, indicating the best relative fit among the candidate models. This specification was therefore selected as the preferred model for subsequent analysis. The automated model selection procedure using the auto.arima() function similarly selected an ARIMA (1, 0, 1) specification, which provided further support for this model choice. As the series was pre-differenced prior to estimation, no further differencing was required within the model (d = 0). The model is, therefore, equivalently expressed as an ARMA (1, 1) specification.

Table 2.

Estimated parameters of the selected ARMA (1, 1) model.

Parameter	Estimate	Std. Error
$\emptyset_1$ (AR1)	0.7653***	0.1208
θ_1 (MA1)	-0.4540***	0.1707
Mean	0.0058***	0.0010

Note. *** $p < 0.01$

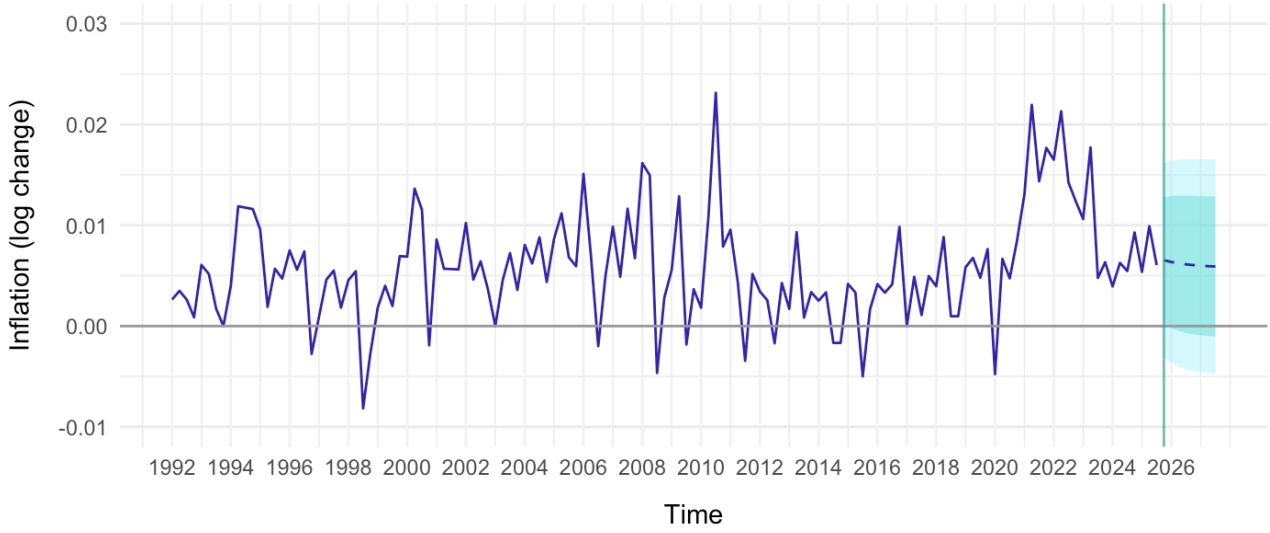
The estimated ARMA (1, 1) model indicates a substantial degree of persistence in inflation dynamics. The autoregressive coefficient ($\emptyset_1$= 0.7653) suggests that inflation in one quarter is strongly influenced by its past values. This also indicates that inflation is persistent in the short-term. The moving average coefficient (θ_1= -0.4540) captures partial short-term adjustment dynamics in the error process. The model also includes a positive mean term (0.0058). This implies that inflation fluctuates around an average level of approximately 0.58% per quarter.

Residual diagnostics (Appendix B) suggest that the selected ARMA (1, 1) model is well-specified. The Ljung–Box test fails to reject the null hypothesis of no autocorrelation (p = 0.7635). This indicates no evidence of residual autocorrelation, consistent with white noise errors. The residual autocorrelation function shows that most autocorrelations lie within the confidence bounds with no systematic autocorrelation. While an isolated spike is observed at higher lags, it did not indicate a persistent structure. The residual distribution also appears approximately normal, further supporting the adequacy of the model.

Next, the fitted model was used to generate short-term forecasts of inflation over the next eight quarters, which are presented in Figure 2.

Figure 2.

Forecast of Quarterly CPI Inflation in New Zealand (ARMA (1, 1)).



Note. Forecasts are generated for an 8-quarter horizon using an ARMA (1, 1) model. Shaded areas represent confidence intervals.

The forecast in Figure 2 suggests a tendency for inflation to revert towards its historical mean. However, the relatively wide confidence intervals reflect increasing forecast uncertainty over the horizon, consistent with the volatility observed in the historical series. This implies that while inflation shocks may occur, the model implies that shocks have diminishing influence over time.

5. Discussion

As the results suggest, inflation in New Zealand exhibits a relatively stable average, with the estimated mean of approximately 2.32% per annum over the sample period. However, the autoregressive component shows high degree of persistence, with past values influencing current inflation. This means that deviation from the baseline may not dissipate quickly.

At the same time, inflation exhibits periods of volatility with sharp fluctuations evident in the historical data (Figures 1 and 2). While shocks do not have permanent effects, as indicated by the moving average component, their impact is only partially reversed in the short term. This suggests that inflation dynamics involve both persistence and gradual adjustment.

In periods of elevated inflation, the estimated dynamics suggest that increases may persist over the short term due to the observed persistence in the series. However, the model indicates that such deviations are unlikely to persist indefinitely, with inflation projected to revert towards

its historical average over time. The forecast results are consistent with this pattern, showing a tendency for inflation to move towards its mean over the forecast horizon, although the wide confidence intervals highlight substantial uncertainty. The increasing uncertainty at longer horizons reflects the underlying volatility of the series.

6. Limitations

The present analysis is based on a univariate ARIMA framework, which relies solely on past values of inflation and does not incorporate exogenous variables. As a result, the model does not explicitly account for external influences, such as geopolitical shocks, supply-side disruptions or policy changes, including fiscal and monetary policy interventions, all of which may influence inflation dynamics.

In addition, the sample period is restricted to 1992–2025, reflecting the introduction of inflation targeting in New Zealand and a shift towards a more stable monetary policy regime. As a result, earlier periods characterised by different inflation dynamics are not included in the analysis. Consequently, the analysis does not capture developments beyond the sample period, including the recent inflationary pressures arising from ongoing global supply-chain disruptions (Breman, 2026). The use of quarterly data also limits the ability to capture higher-frequency dynamics and potential seasonal patterns. While no clear seasonal structure was identified, more granular data may reveal short-term fluctuations.

Finally, while the ARIMA model provides a useful benchmark for describing underlying inflation dynamics, it does not capture causal relationships. The results should therefore be interpreted as a baseline representation of inflation behaviour in New Zealand, against which more complex models incorporating exogenous drivers or machine learning approaches may be evaluated.

References

Ball, C., Brunton, M., Chipeniuk, K., & Jacob, P. (2025, June). Reflections on 35 years of flexible inflation targeting: Lessons from the RBNZ Research Conference. *Reserve Bank Bulletin, 88*(5), 1–15.

Box, G. E. P., Jenkins, G. M., Reinsel, G. C., & Ljung, G. M. (2016). *Time series analysis: Forecasting and control* (5th ed.). John Wiley & Sons.

Breman, A. (2026, March 24). *Global shockwaves to Kiwi shores: The impact of the Iran conflict on New Zealand* [Speech transcript]. Reserve Bank of New Zealand. https://www.rbnz.govt.nz/-/media/project/sites/rbnz/files/publications/speeches/2026/global-shockwaves-to-kiwi-shores.pdf

Buckle, R. A. (2018, December). *Thirty years of inflation targeting in New Zealand: The origins, evolution and influence of a monetary policy innovation* (Working Paper No. 14/2018). https://ir.wgtn.ac.nz/items/c10882d8-8867-4c1c-b0e7-62dedd70a56d

Dickey, D. A., & Fuller, W. A. (1979). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association, 74*(366), 427–431. https://doi.org/10.1080/01621459.1979.10482531

Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: principles and practice*. OTexts.

McDermott, J., & Williams, R. (2018, April 12). *Inflation targeting in New Zealand: An experience in evolution* [Conference session]. Reserve Bank of Australia Conference on Central Bank Frameworks, Sydney, Australia.

Stats NZ. (2026). *Consumers price index (CPI)*. https://www.stats.govt.nz/indicators/consumers-price-index-cpi/

Appendix A: R Script

```
library(tidyverse)
library(tseries)
library(forecast)

df <- consumers_price_index_december_2025_quarter_index_numbers

glimpse(df)
summary(df)

#Inspect Groups for CPI
df %>% distinct(Group)

#Missing or zero values check
missing_data_values <- df %>%
  filter(is.na(Data_value) | Data_value == 0)

#Filter CPI to All Groups
df_clean <- df %>%
  filter(Group == "CPI All Groups for New Zealand") %>%
  select(Period, Data_value) %>%
  mutate(
    Period = as.character(Period),
    Year = substr(Period, 1, 4),
    Month = substr(Period, 6, 7),
    Date = as.Date(paste0(Year, "-", Month, "-01"))
  ) %>%
  arrange(Date)

#Filter sample period to 1992-2025
df_CPI <- df_clean %>%
  filter(
    Date >= as.Date("1992-03-01"),
    Date <= as.Date("2025-12-01")
  ) %>%
  select(Data_value, Date)

#Log differencing CPI
df_CPI <- df_CPI %>%
  mutate(
    log_CPI = log(Data_value),
    pi_t = log_CPI - lag(log_CPI)
  ) %>%
  filter(!is.na(pi_t))

#Quarterly Inflation Historical Plot
ggplot(df_CPI, aes(x = Date, y = pi_t)) +
  geom_line(color = "#3626A7", size = 0.5) +
  geom_hline(yintercept = 0, color = "grey60", size = 0.3) +
  scale_x_date(
    limits = c(as.Date("1992-06-01"), as.Date("2025-12-01")),
    breaks = seq(as.Date("1992-06-01"), as.Date("2024-06-01"), by = "2 years"),
    date_labels = "%Y"
  ) +
  scale_y_continuous(
    limits = c(-0.01, 0.03),
    breaks = seq(-0.01, 0.03, by = 0.01)
  ) +
  labs(x = "Time", y = "Inflation (log change)") +
  theme_minimal() +
  theme(
    axis.title.y = element_text(size = 11, margin = margin(r = 10)),
    axis.title.x = element_text(size = 11, margin = margin(t = 10))
  )

#Quarterly time series object
inflation_ts <- ts(df_CPI$pi_t, start = c(1992, 2), frequency = 4)

# Augmented Dickey-Fuller (ADF) test for Stationarity
adf.test(inflation_ts)

#ACF and PACF Plots
acf(inflation_ts, main = "ACF of Quarterly Inflation")
pacf(inflation_ts, main = "PACF of Quarterly Inflation")
```

```
#Candidate ARIMA models
m_ar1 <- Arima(inflation_ts, order = c(1, 0, 0))
m_ma1 <- Arima(inflation_ts, order = c(0, 0, 1))
m_arma11 <- Arima(inflation_ts, order = c(1, 0, 1))
m_arma21 <- Arima(inflation_ts, order = c(2, 0, 1))

#Auto.arima Selection
m_auto <- auto.arima(inflation_ts, d = 0, seasonal = FALSE)

#Model comparison
model_compare <- tibble(
  Model = c("AR(1)", "MA(1)", "ARMA(1,1)", "ARMA(2,1)", "AUTO"),
  AIC = c(AIC(m_ar1), AIC(m_ma1), AIC(m_arma11), AIC(m_arma21), AIC(m_auto)),
  BIC = c(BIC(m_ar1), BIC(m_ma1), BIC(m_arma11), BIC(m_arma21), BIC(m_auto))
)
model_compare

#Residual diagnostics
checkresiduals(m_ar1)
checkresiduals(m_arma11)

#Select preferred model
m_best <- m_arma11

#Forecast
fc_best <- forecast(m_best, h = 8)

#Forecast data frame for plotting
fc_df <- data.frame(
  Date = seq(as.Date("2026-03-01"), by = "quarter", length.out = 8),
  mean = as.numeric(fc_best$mean),
  lower80 = as.numeric(fc_best$lower[, 1]),
  upper80 = as.numeric(fc_best$upper[, 1]),
  lower95 = as.numeric(fc_best$lower[, 2]),
  upper95 = as.numeric(fc_best$upper[, 2])
)

#Forecast plot
ggplot() +
  geom_line(data = df_CPI, aes(x = Date, y = pi_t), color = "#3626A7", size = 0.5) +
  geom_ribbon(data = fc_df, aes(x = Date, ymin = lower95, ymax = upper95),
              fill = "#8EEDF7", alpha = 0.4) +
  geom_ribbon(data = fc_df, aes(x = Date, ymin = lower80, ymax = upper80),
              fill = "#78E0DC", alpha = 0.6) +
  geom_line(data = fc_df, aes(x = Date, y = mean),
            color = "#3626A7", size = 0.5, linetype = "dashed") +
  geom_hline(yintercept = 0, color = "grey60", size = 0.5) +
  geom_vline(xintercept = as.Date("2026-03-01"),
             linetype = "solid", color = "#2A9D8F", size = 0.5, alpha = 0.7) +
  scale_x_date(
    limits = c(as.Date("1992-06-01"), as.Date("2027-12-01")),
    breaks = seq(as.Date("1992-06-01"), as.Date("2026-06-01"), by = "2 years"),
    date_labels = "%Y"
  ) +
  scale_y_continuous(
    limits = c(-0.01, 0.03),
    breaks = seq(-0.01, 0.03, by = 0.01)
  ) +
  labs(x = "Time", y = "Inflation (log change)") +
  theme_minimal() +
  theme(
    axis.title.y = element_text(size = 11, margin = margin(r = 10)),
    axis.title.x = element_text(size = 11, margin = margin(t = 10)),
    panel.grid.major.x = element_line(color = "grey95")
  )

summary(m_arma11)
```

Appendix B: Model Diagnostics

Figure B1: ACF of Inflation

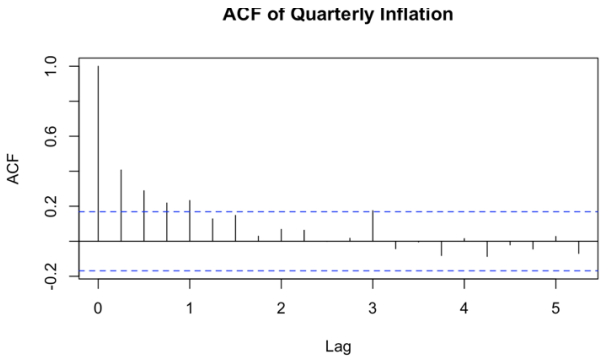


Figure B2: PACF of Inflation

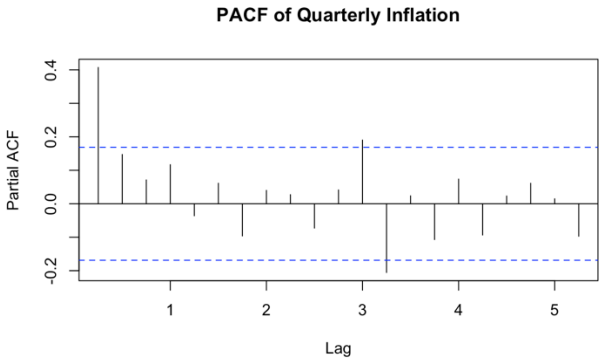
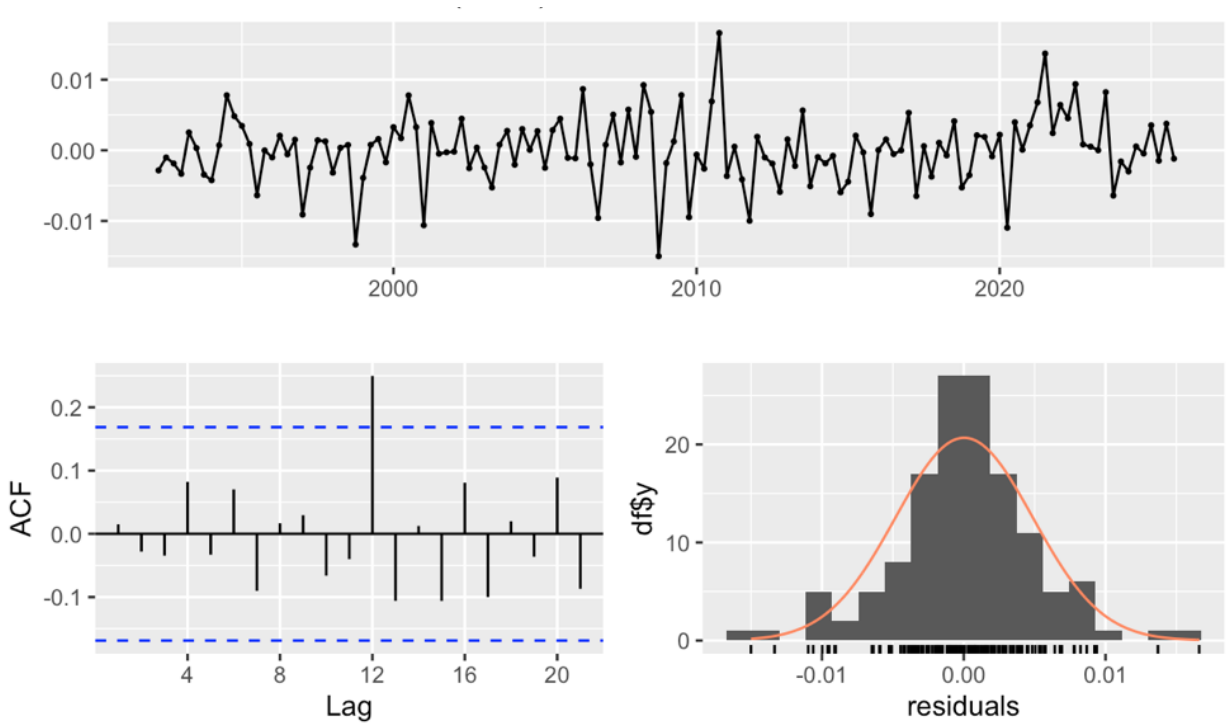



Figure B3: Residual diagnostics of ARIMA (1, 0, 1)